

SOLUTION

1. Doctor Shift Assignment using Backtracking

Step 1:

Doctors = D1, D2, D3

Shifts = Morning, Afternoon, Night

Step 2:

Assign D1 → Morning ✓

Assign D2 → Afternoon ✓

Assign D3 → Night ✓

Constraint Checks:

- D1 cannot work Night ✓
- D2 works before D3 (Afternoon < Night) ✓
- D3 cannot work Morning ✓
- Only one doctor per shift ✓
- All doctors assigned exactly one shift ✓

Final Schedule:

Morning → D1

Afternoon → D2

Night → D3

2. Robot Path Finding using BFS

Step 1:

Start at S.

Step 2:

Expand nodes level by level using BFS.

Step 3:

Avoid obstacle cells and visit only valid neighboring cells.

Step 4:

Reach the Goal (G).

Optimal Path:

S → Right → Right → Down → Down → Right → Right → Down → G

Total Moves = 8
Cost per Move = 1
Total Cost = 8

3. Autonomous Rescue Robot

a) Constraint Analysis:

- Obstacles cannot be crossed.
- Robot can sense only adjacent cells.
- Normal move cost = 1.
- Risky cell adds an extra cost of 2.
- Robot updates its knowledge while moving.

b) Solution Approach (Uniform Cost Search / Online Search):

1. Start from S.
2. Observe neighboring cells.
3. Calculate movement cost.
4. Avoid obstacles.
5. Choose the lowest-cost path.
6. Update knowledge after each move.
7. Continue until G is reached.

c) AI Concepts:

- Intelligent Agent – Senses and acts autonomously.
- Rationality – Chooses the safest and lowest-cost path.
- Environment Type – Partially observable and dynamic.

d) Justification:

Uniform Cost Search (or Online Search) is suitable because it finds the minimum-cost path, avoids obstacles, handles risky cells, and updates decisions dynamically.

Result:

The rescue robot safely reaches the survivor with the minimum total cost.